BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: KT-XI-23-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 20 | **Paper Code:** Code B

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. If A = [1 3 1; -1 2 0; 2 0 -2] then A_{23} = ...

(A) 0 (B) -6

(C) 6 (D) 1

Qii. The transpose of a row matrix is .....

(A) column matrix (B) Row matrix

(C) Square matrix (D) Rectangular matrix

Qiii. If α, β and γ are direction angles of a vector r = xi + yj + zk, then cos γ = ...

(A) x/|r| (B) y/|r|

(C) z/|r| (D) None of these

Qiv. Find the fifth term of the sequence, -8,-3,2,7, ...

(A) -10 (B) 12

(C) -14 (D) 16

Qv. The numbers in the range of a sequence are real numbers.

(A) terms of the sequence (B) Domain

(C) Series (D) Real number

Qvi. $\Sigma(k=0 \text{ to } 3)\ 2^k$ = ____

(A) 13 (B) 14

(C) 15 (D) 16

Qvii. If det (A) = 5, then find det (15A) where A is of order 2 × 2.

(A) 225 (B) 75

(C) 375 (D) 1125

Qviii. The geometric mean of 9 and 16 is .....

(A) 12 (B) 9

(C) 16 (D) 7

Qix. P(E) = n(E)/n(S)

(A) 0 > P(E) > 1 (B) 0 < P(E) < 1

(C) 0 ≤ P(E) ≤ 1 (D) 0 < P(E) > 1

Qx. The maximum distance between the graph of the Sine or Cosine and horizontal axis is called ....

(A) Amplitude (B) Frequency

(C) Period (D) None of these

Qxi. Maximum value of y = 3+2 Cos (3θ – 2)

(A) 1 (B) 2

(C) 3 (D) 5

Qxii. $i^8 + i^{19}$ = ...

(A) 0 (B) 1

(C) i (D) -i

Qxiii. If $z_1 = 2+4i$ and $z_2 = 1-3i$ then $z_1 - 3z_2$ = ...

(A) 1 – 3i (B) –1 + 13i

(C) 5+13i (D) –5 + 13i

Qxiv. ${}^{10}C_1 + {}^{10}C_3 + {}^{10}C_5 + ...+ {}^{10}C_9$ = ...

(A) 512 (B) 1024

(C) 2048 (D) 1023

Qxv. If $f(x) = x^2 - 3x + 4$ then find the values of x satisfying the equation, f(x) = f(2x + 1)

(A) 5/3 (B) 2/3

(C) 1 (D) 1/3

Qxvi. The variables used in the system of linear inequalities relating to the problem are non-negative called ....

(A) Decision variables (B) problem constraints

(C) linear programming (D) feasible solution

Qxvii. The angles which are connected with basic angles of measure θ by a right angle or its multiple are known as .....

(A) trigonometric ratio
(B) allied angles
(C) not allied angles
(D) law of sine

Qxviii. Cos(2 π +θ) = ...

(A) Sin θ
(B) – Sin θ
(C) Cos θ
(D) – Cos θ

Qxix. Cos (α + β) = ...

(A) cosαcosβ–SinaSinβ
(B) SinaSinβ+CosaCosβ
(C) CosaSinβ+CosβSinα
(D) CosaCosβ+SinaSinβ

Qxx. $1/2\ b^2 \sin\alpha \sin\beta / \sin\theta$

(A) Δ
(B) Δ^2
(C) a^2
(D) $(1-n)^n$

Section "B"

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 50

QQ.2. Attempt any TEN parts. All parts carry equal marks.

i Express in standard form a+ib: (1+2i)/(2+i) + (4-i)/(3+2i)

ii If z1 = 2+3i and z2 = 3-3i, verify that conjugate(z1+z2) = conjugate(z1) + conjugate(z2)

iii Find the value of a and b if the matrix is symmetric. [0 2b -2; 3 1 3; 3a 3 -1]

iv Solve for x, if |x 2 3; 0 -1 1; 0 4 5| = 9

v Find a unit vector that is perpendicular to a = i – 2j + 3k and b = 2i + j – k

vi If A + B + C = 0 then show that A × B = B × C = C × A

vii Find the 2nd term of an H.P is 1/2 and 5th term is -1/4. Find the 12th term.

viii If for G.P. $a_{10} = l$, $a_{13} = m$, $a_{16} = n$ then show that $l.n = m^2$

ix Find the sum of nth terms of the series whose nth term is 2^(n-1) + $8n^3 - 6n^2$

x Find n. when ${}^{2n}C_3 : {}^{n}C_2 = 36:3$

xi If $f(x) = x^3 - 2$ find $f^{-1}(3)$

xii Maximize f(x,y) = 2x + y subject to the constraints, x+y ≤ 6, x+y ≥1, x,y ≥ 0

xiii Show that $2\tan^{-1} A = \text{Tan}^{-1}(2A/(1-A^2))$

Section "C"

Maximum Marks: 30

Note: Note: Attempt any THREE questions. All questions carry equal marks.

QQ.3.

a find the middle term in the expansion of $(a/x + bx)^8$

b Show that $\text{Sin}^4\theta = 3/8 - 1/2 \text{ Cos } 2\theta + 1/8 \text{ Cos}4\theta$

QQ.4.

a Find the area of the triangle ABC in which b=14, c=12, $\alpha = 82°$

b Prove that $\text{Sin}(\alpha+\beta) . \text{Sin } (\alpha - \beta) = \text{Cos}^2\beta - \text{Cos}^2\alpha$

QQ.5.

a $2^7 xy^3$ is a term in the expansion of $(ax + 2y)^4$. Find a.

b Solve the equation $2\text{Sin}^{-1}x - \text{Cos}^{-1}x = \pi/2$

QQ.6.

a For what value of n, (a^(n+1) + b^(n+1))/(a^n + b^n) is arithmetic mean between a and b

b show that the vectors 3i – 2j + k, i - 3j + 5k and 2i + j – 4k form a right triangle.